LAPORAN TUGAS BESAR PRAKTIKUM DP 5

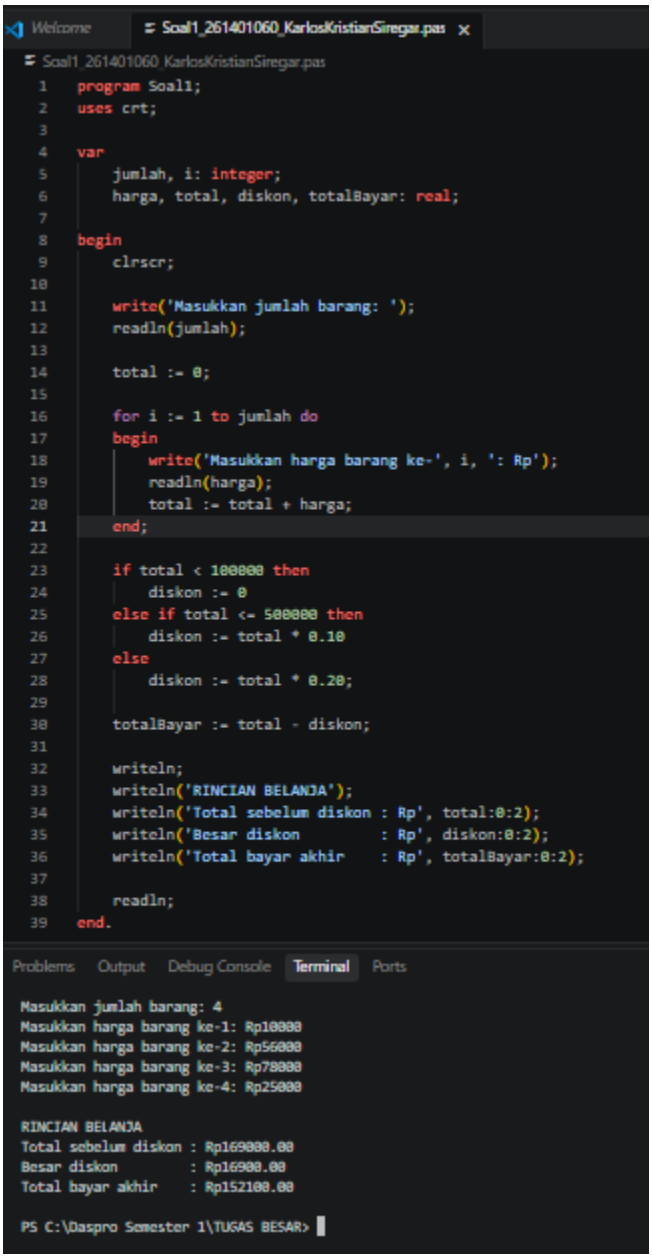


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LAB : DP 5

[Soal 1]



```
Welcome    Soal1_261401060_KarlosKristianSiregar.pas

Soal1_261401060_KarlosKristianSiregar.pas
 1  program Soal1;
 2  uses crt;
 3
 4  var
 5      jumlah, i: integer;
 6      harga, total, diskon, totalBayar: real;
 7
 8  begin
 9      clrscr;
10
11      write('Masukkan jumlah barang: ');
12      readln(jumlah);
13
14      total := 0;
15
16      for i := 1 to jumlah do
17      begin
18          write('Masukkan harga barang ke-', i, ': Rp');
19          readln(harga);
20          total := total + harga;
21      end;
22
23      if total < 100000 then
24          diskon := 0
25      else if total <= 500000 then
26          diskon := total * 0.10
27      else
28          diskon := total * 0.20;
29
30      totalBayar := total - diskon;
31
32      writeln;
33      writeln('RINCIAN BELANJA');
34      writeln('Total sebelum diskon : Rp', total:0:2);
35      writeln('Besar diskon         : Rp', diskon:0:2);
36      writeln('Total bayar akhir    : Rp', totalBayar:0:2);
37
38      readln;
39  end.

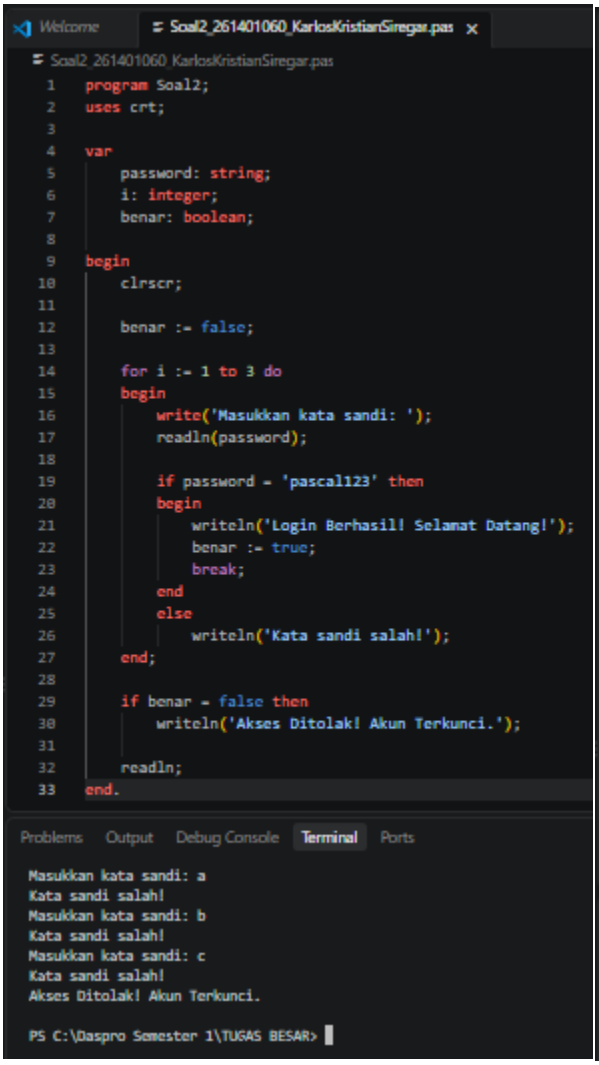
Problems  Output  Debug Console  Terminal  Ports

Masukkan jumlah barang: 4
Masukkan harga barang ke-1: Rp10000
Masukkan harga barang ke-2: Rp56000
Masukkan harga barang ke-3: Rp78000
Masukkan harga barang ke-4: Rp25000

RINCIAN BELANJA
Total sebelum diskon : Rp169000.00
Besar diskon         : Rp16900.00
Total bayar akhir    : Rp152100.00

PS C:\Daspro Semester 1\TUGAS BESAR>
```

[Soal 2]



```
Welcome | Soal2_261401060_KarlosKristianSiregar.pas

Soal2_261401060_KarlosKristianSiregar.pas
1   program Soal2;
2   uses crt;
3
4   var
5       password: string;
6       i: integer;
7       benar: boolean;
8
9   begin
10      clrscr;
11
12      benar := false;
13
14      for i := 1 to 3 do
15      begin
16          write('Masukkan kata sandi: ');
17          readln(password);
18
19          if password = 'pascal123' then
20          begin
21              writeln('Login Berhasil! Selamat Datang!');
22              benar := true;
23              break;
24          end
25          else
26              writeln('Kata sandi salah!');
27      end;
28
29      if benar = false then
30          writeln('Akses Ditolak! Akun Terkunci.');
31
32      readln;
33  end.

Problems  Output  Debug Console  Terminal  Ports

Masukkan kata sandi: a
Kata sandi salah!
Masukkan kata sandi: b
Kata sandi salah!
Masukkan kata sandi: c
Kata sandi salah!
Akses Ditolak! Akun Terkunci.

PS C:\Daspro Semester 1\TUGAS BESAR>
```



```
Welcome | Soal2_261401060_KarlosKristianSiregar.pas

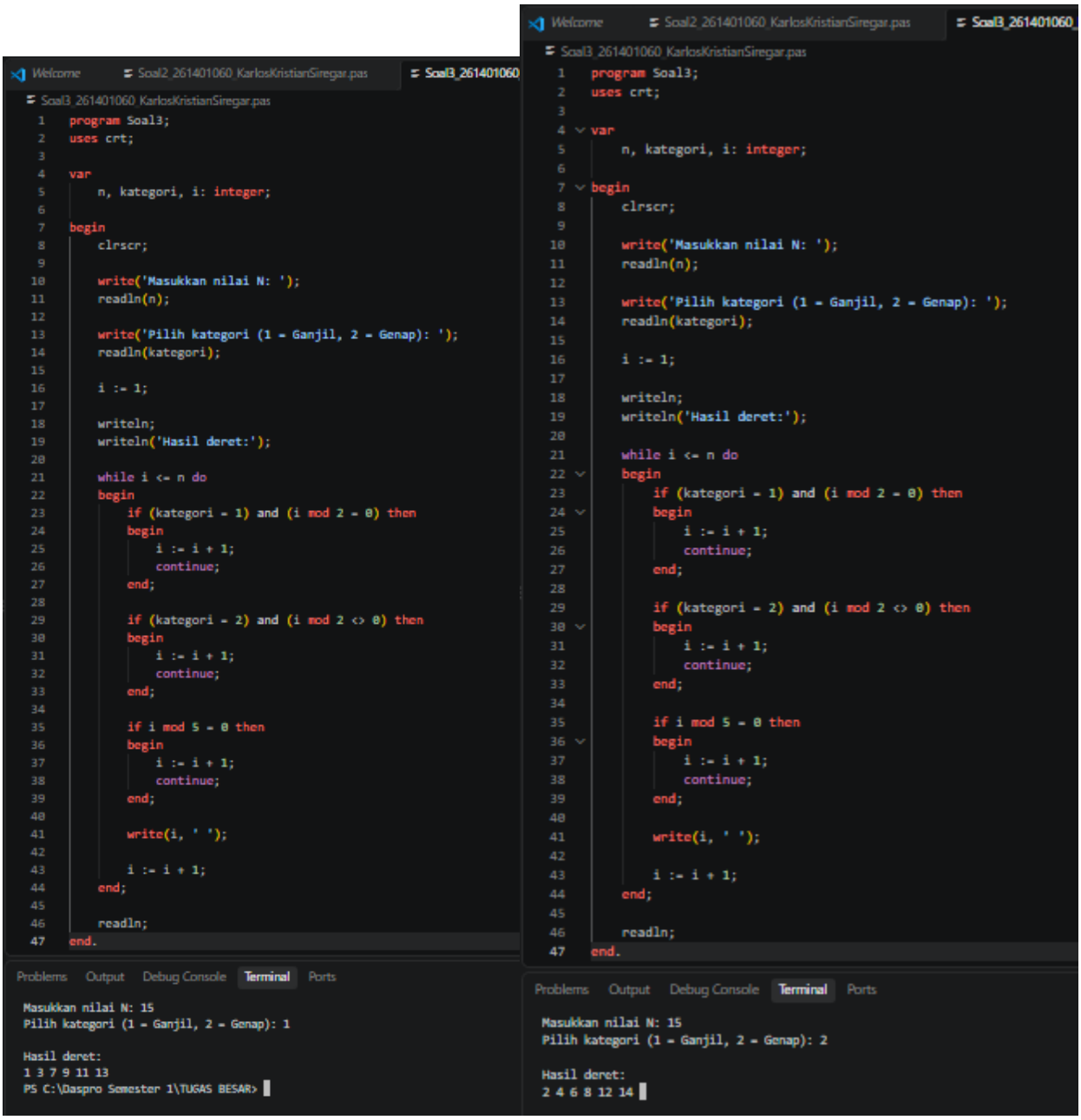
Soal2_261401060_KarlosKristianSiregar.pas
1   program Soal2;
2   uses crt;
3
4   var
5       password: string;
6       i: integer;
7       benar: boolean;
8
9   begin
10      clrscr;
11
12      benar := false;
13
14      for i := 1 to 3 do
15      begin
16          write('Masukkan kata sandi: ');
17          readln(password);
18
19          if password = 'pascal123' then
20          begin
21              writeln('Login Berhasil! Selamat Datang!');
22              benar := true;
23              break;
24          end
25          else
26              writeln('Kata sandi salah!');
27      end;
28
29      if benar = false then
30          writeln('Akses Ditolak! Akun Terkunci.');
31
32      readln;
33  end.

Problems  Output  Debug Console  Terminal  Ports

Masukkan kata sandi: pascal123
Login Berhasil! Selamat Datang!

PS C:\Daspro Semester 1\TUGAS BESAR>
```

[Soal3]



Welcome | Soal2_261401060_KarlosKristianSiregar.pas | Soal3_261401060_KarlosKristianSiregar.pas

```
program Soal3;
uses crt;

var
    n, kategori, i: integer;

begin
    clrscr;

    write('Masukkan nilai N: ');
    readln(n);

    write('Pilih kategori (1 = Ganjil, 2 = Genap): ');
    readln(kategori);

    i := 1;

    writeln;
    writeln('Hasil deret:');

    while i <= n do
    begin
        if (kategori = 1) and (i mod 2 = 0) then
        begin
            i := i + 1;
            continue;
        end;

        if (kategori = 2) and (i mod 2 <> 0) then
        begin
            i := i + 1;
            continue;
        end;

        if i mod 5 = 0 then
        begin
            i := i + 1;
            continue;
        end;

        write(i, ' ');

        i := i + 1;
    end;

    readln;
end.
```

Problems | Output | Debug Console | Terminal | Ports

```
Masukkan nilai N: 15
Pilih kategori (1 = Ganjil, 2 = Genap): 1

Hasil deret:
1 3 7 9 11 13
PS C:\Daspro Semester 1\TUGAS BESAR>
```

Welcome | Soal2_261401060_KarlosKristianSiregar.pas | Soal3_261401060_KarlosKristianSiregar.pas

```
program Soal3;
uses crt;

var
    n, kategori, i: integer;

begin
    clrscr;

    write('Masukkan nilai N: ');
    readln(n);

    write('Pilih kategori (1 = Ganjil, 2 = Genap): ');
    readln(kategori);

    i := 1;

    writeln;
    writeln('Hasil deret:');

    while i <= n do
    begin
        if (kategori = 1) and (i mod 2 = 0) then
        begin
            i := i + 1;
            continue;
        end;

        if (kategori = 2) and (i mod 2 <> 0) then
        begin
            i := i + 1;
            continue;
        end;

        if i mod 5 = 0 then
        begin
            i := i + 1;
            continue;
        end;

        write(i, ' ');

        i := i + 1;
    end;

    readln;
end.
```

Problems | Output | Debug Console | Terminal | Ports

```
Masukkan nilai N: 15
Pilih kategori (1 = Ganjil, 2 = Genap): 2

Hasil deret:
2 4 6 8 12 14
```

[Soal 4]



```
program Soal4;
uses crt;

var
    pilihan: integer;
    a, b, hasil: real;
    ulang: char;

begin
    clrscr;

    repeat
        writeln('===== KALKULATOR =====');
        writeln('1. Penjumlahan');
        writeln('2. Pengurangan');
        writeln('3. Perkalian');
        writeln('4. Pembagian Real');
        writeln('5. DIV');
        writeln('6. MOD');
        writeln;

        write('Pilih operasi: ');
        readln(pilihan);

        write('Masukkan angka pertama: ');
        readln(a);

        write('Masukkan angka kedua: ');
        readln(b);

        case pilihan of
            1:
                begin
                    hasil := a + b;
                    writeln('Hasil = ', hasil:0:2);
                end;

            2:
                begin
                    hasil := a - b;
                    writeln('Hasil = ', hasil:0:2);
                end;

            3:
                begin
                    hasil := a * b;
                    writeln('Hasil = ', hasil:0:2);
                end;

            4:
                begin
                    if b <> 0 then
                    begin
```

```
===== KALKULATOR =====
1. Penjumlahan
2. Pengurangan
3. Perkalian
4. Pembagian Real
5. DIV
6. MOD

Pilih operasi: 3
Masukkan angka pertama: 7
Masukkan angka kedua: 5
Hasil = 35.00

Apakah ingin melakukan perhitungan lagi? (Y/T): Y
```



```
begin
    repeat
        case pilihan of
            1:
                begin
                end;

            2:
                begin
                    hasil := a - b;
                    writeln('Hasil = ', hasil:0:2);
                end;

            3:
                begin
                    hasil := a * b;
                    writeln('Hasil = ', hasil:0:2);
                end;

            4:
                begin
                    if b <> 0 then
                    begin
                        hasil := a / b;
                        writeln('Hasil = ', hasil:0:2);
                    end
                    else
                        writeln('Tidak bisa dibagi 0');
                end;

            5:
                writeln('Hasil DIV = ', trunc(a) div trunc(b));

            6:
                writeln('Hasil MOD = ', trunc(a) mod trunc(b));

            else
                writeln('Pilihan tidak tersedia');
        end;

        writeln;
        write('Apakah ingin melakukan perhitungan lagi? (Y/T): ');
        readln(ulang);

        writeln;

    until (ulang = 'T') or (ulang = 't');

    writeln('Program selesai.');

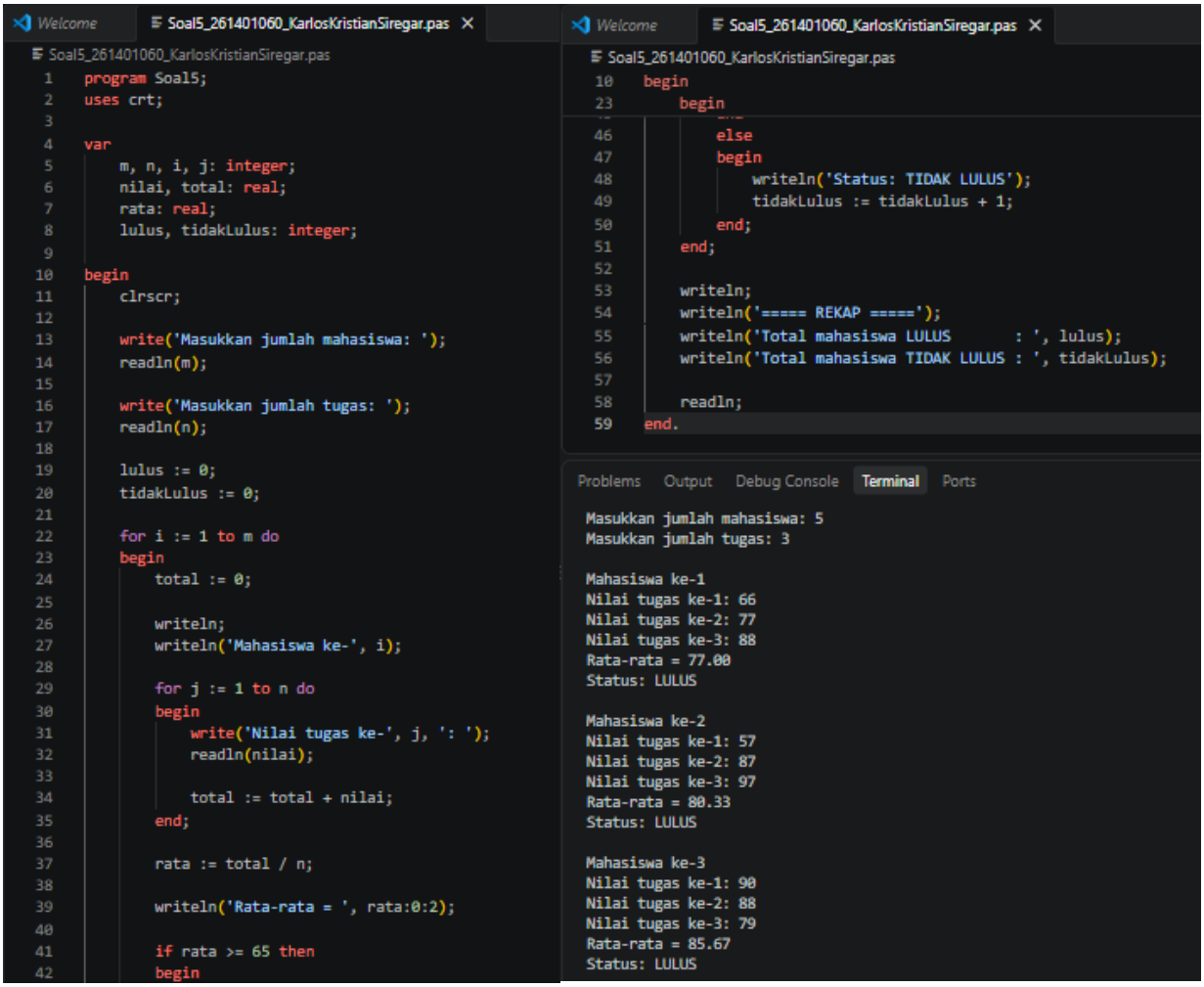
    readln;
end.
```

```
===== KALKULATOR =====
1. Penjumlahan
2. Pengurangan
3. Perkalian
4. Pembagian Real
5. DIV
6. MOD

Pilih operasi: 3
Masukkan angka pertama: 7
Masukkan angka kedua: 5
Hasil = 35.00

Apakah ingin melakukan perhitungan lagi? (Y/T): Y
```

[Soal 5]



Welcome | Soal5_261401060_KarlosKristianSiregar.pas

Soal5_261401060_KarlosKristianSiregar.pas

```
1   program Soal5;
2   uses crt;
3
4   var
5       m, n, i, j: integer;
6       nilai, total: real;
7       rata: real;
8       lulus, tidakLulus: integer;
9
10  begin
11      clrscr;
12
13      write('Masukkan jumlah mahasiswa: ');
14      readln(m);
15
16      write('Masukkan jumlah tugas: ');
17      readln(n);
18
19      lulus := 0;
20      tidakLulus := 0;
21
22      for i := 1 to m do
23      begin
24          total := 0;
25
26          writeln;
27          writeln('Mahasiswa ke-', i);
28
29          for j := 1 to n do
30          begin
31              write('Nilai tugas ke-', j, ': ');
32              readln(nilai);
33
34              total := total + nilai;
35          end;
36
37          rata := total / n;
38
39          writeln('Rata-rata = ', rata:0:2);
40
41          if rata >= 65 then
42          begin
```

Welcome | Soal5_261401060_KarlosKristianSiregar.pas

Soal5_261401060_KarlosKristianSiregar.pas

```
10  begin
23      begin
46          else
47          begin
48              writeln('Status: TIDAK LULUS');
49              tidakLulus := tidakLulus + 1;
50          end;
51      end;
52
53      writeln;
54      writeln('===== REKAP =====');
55      writeln('Total mahasiswa LULUS       : ', lulus);
56      writeln('Total mahasiswa TIDAK LULUS : ', tidakLulus);
57
58      readln;
59  end.
```

Problems | Output | Debug Console | Terminal | Ports

```
Masukkan jumlah mahasiswa: 5
Masukkan jumlah tugas: 3

Mahasiswa ke-1
Nilai tugas ke-1: 66
Nilai tugas ke-2: 77
Nilai tugas ke-3: 88
Rata-rata = 77.00
Status: LULUS

Mahasiswa ke-2
Nilai tugas ke-1: 57
Nilai tugas ke-2: 87
Nilai tugas ke-3: 97
Rata-rata = 80.33
Status: LULUS

Mahasiswa ke-3
Nilai tugas ke-1: 90
Nilai tugas ke-2: 88
Nilai tugas ke-3: 79
Rata-rata = 85.67
Status: LULUS
```

Welcome | Soal5_261401060_KarlosKristianSiregar.pas

Soal5_261401060_KarlosKristianSiregar.pas

```
10   begin
23       begin
46           else
47           begin
48               writeln('Status: TIDAK LULUS');
49               tidakLulus := tidakLulus + 1;
50           end;
51       end;
52
53       writeln;
54       writeln('===== REKAP =====');
55       writeln('Total mahasiswa LULUS       : ', lulus);
56       writeln('Total mahasiswa TIDAK LULUS : ', tidakLulus);
57
58       readln;
59   end.
```

Problems | Output | Debug Console | Terminal | Ports

```
Nilai tugas ke-3: 97
Rata-rata = 80.33
Status: LULUS

Mahasiswa ke-3
Nilai tugas ke-1: 90
Nilai tugas ke-2: 88
Nilai tugas ke-3: 79
Rata-rata = 85.67
Status: LULUS

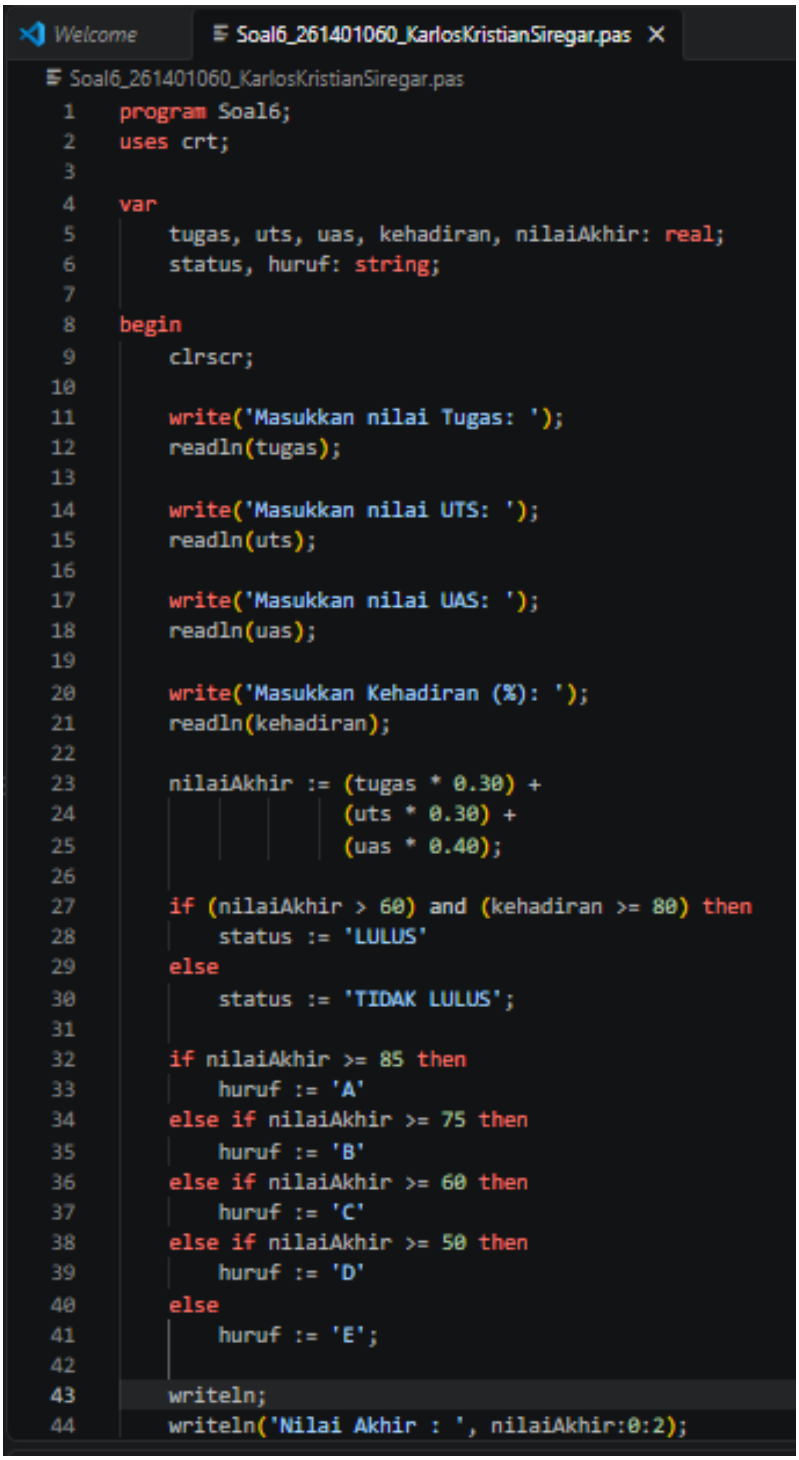
Mahasiswa ke-4
Nilai tugas ke-1: 77
Nilai tugas ke-2: 98
Nilai tugas ke-3: 86
Rata-rata = 87.00
Status: LULUS

Mahasiswa ke-5
Nilai tugas ke-1: 78
Nilai tugas ke-2: 97
Nilai tugas ke-3: 68
Rata-rata = 81.00
Status: LULUS

===== REKAP =====
Total mahasiswa LULUS       : 5
Total mahasiswa TIDAK LULUS : 0

PS C:\Daspro Semester 1\TUGAS BESAR>
```

[Soal 6]



Welcome | Soal6_261401060_KarlosKristianSiregar.pas

Soal6_261401060_KarlosKristianSiregar.pas

```
1   program Soal6;
2   uses crt;
3
4   var
5       tugas, uts, uas, kehadiran, nilaiAkhir: real;
6       status, huruf: string;
7
8   begin
9       clrscr;
10
11      write('Masukkan nilai Tugas: ');
12      readln(tugas);
13
14      write('Masukkan nilai UTS: ');
15      readln(uts);
16
17      write('Masukkan nilai UAS: ');
18      readln(uas);
19
20      write('Masukkan Kehadiran (%): ');
21      readln(kehadiran);
22
23      nilaiAkhir := (tugas * 0.30) +
24                    (uts * 0.30) +
25                    (uas * 0.40);
26
27      if (nilaiAkhir > 60) and (kehadiran >= 80) then
28          status := 'LULUS'
29      else
30          status := 'TIDAK LULUS';
31
32      if nilaiAkhir >= 85 then
33          huruf := 'A'
34      else if nilaiAkhir >= 75 then
35          huruf := 'B'
36      else if nilaiAkhir >= 60 then
37          huruf := 'C'
38      else if nilaiAkhir >= 50 then
39          huruf := 'D'
40      else
41          huruf := 'E';
42
43      writeln;
44      writeln('Nilai Akhir : ', nilaiAkhir:0:2);
```

Welcome | Soal6_261401060_KarlosKristianSiregar.pas

Soal6_261401060_KarlosKristianSiregar.pas

```
 8   begin
38      else if nilaiAkhir >= 50 then
39          huruf := 'D'
40      else
41          huruf := 'E';
42
43      writeln;
44      writeln('Nilai Akhir : ', nilaiAkhir:0:2);
45      writeln('Nilai Huruf : ', huruf);
46      writeln('Status      : ', status);
47
48      readln;
49  end.
```

Problems | Output | Debug Console | Terminal | Ports

```
Masukkan nilai Tugas: 88
Masukkan nilai UTS: 76
Masukkan nilai UAS: 97
Masukkan Kehadiran (%): 90

Nilai Akhir : 88.00
Nilai Huruf : A
Status      : LULUS

PS C:\Daspro Semester 1\TUGAS BESAR>
```

Welcome | Soal6_261401060_KarlosKristianSiregar.pas

Soal6_261401060_KarlosKristianSiregar.pas

```
 8   begin
38      else if nilaiAkhir >= 50 then
39          huruf := 'D'
40      else
41          huruf := 'E';
42
43      writeln;
44      writeln('Nilai Akhir : ', nilaiAkhir:0:2);
45      writeln('Nilai Huruf : ', huruf);
46      writeln('Status      : ', status);
47
48      readln;
49  end.
```

Problems | Output | Debug Console | Terminal | Ports

```
Masukkan nilai Tugas: 88
Masukkan nilai UTS: 98
Masukkan nilai UAS: 78
Masukkan Kehadiran (%): 60

Nilai Akhir : 87.00
Nilai Huruf : A
Status      : TIDAK LULUS

PS C:\Daspro Semester 1\TUGAS BESAR>
```

[Soal 7]

Soal7_261401060_KarlosKristianSiregar.pas

```
program Soal7;
uses crt;

var
    kode: char;
    jam, tarif: integer;

begin
    clrscr;

    write('Masukkan kode kendaraan (M/K/B): ');
    readln(kode);

    write('Masukkan lama parkir (jam): ');
    readln(jam);

    case kode of

        'M', 'm':
        begin
            if jam <= 1 then
                tarif := 2000
            else
                tarif := 2000 + (jam - 1) * 3000;
        end;

        'K', 'k':
        begin
            if jam <= 1 then
                tarif := 2000
            else
                tarif := 2000 + (jam - 1) * 1000;
        end;

        'B', 'b':
        begin
            if jam <= 1 then
                tarif := 10000
            else
                tarif := 10000 + (jam - 1) * 5000;
        end;

        else
        begin
            writeln('Kode kendaraan tidak valid');
            readln;
            exit;
```

Soal7_261401060_KarlosKristianSiregar.pas

```
Soal7_261401060_KarlosKristianSiregar.pas
 8   begin
17       case kode of
44           begin
47               exit;
48           end;
49       end;
50
51       if (kode = 'M') or (kode = 'm') then
52       begin
53           if tarif > 30000 then
54               tarif := 30000;
55       end
56       else if (kode = 'K') or (kode = 'k') then
57       begin
```

Problems Output Debug Console **Terminal** Ports

```
Masukkan kode kendaraan (M/K/B): B
Masukkan lama parkir (jam): 5
Total tarif parkir = Rp30000

PS C:\Daspro Semester 1\TUGAS BESAR>
```

Soal7_261401060_KarlosKristianSiregar.pas

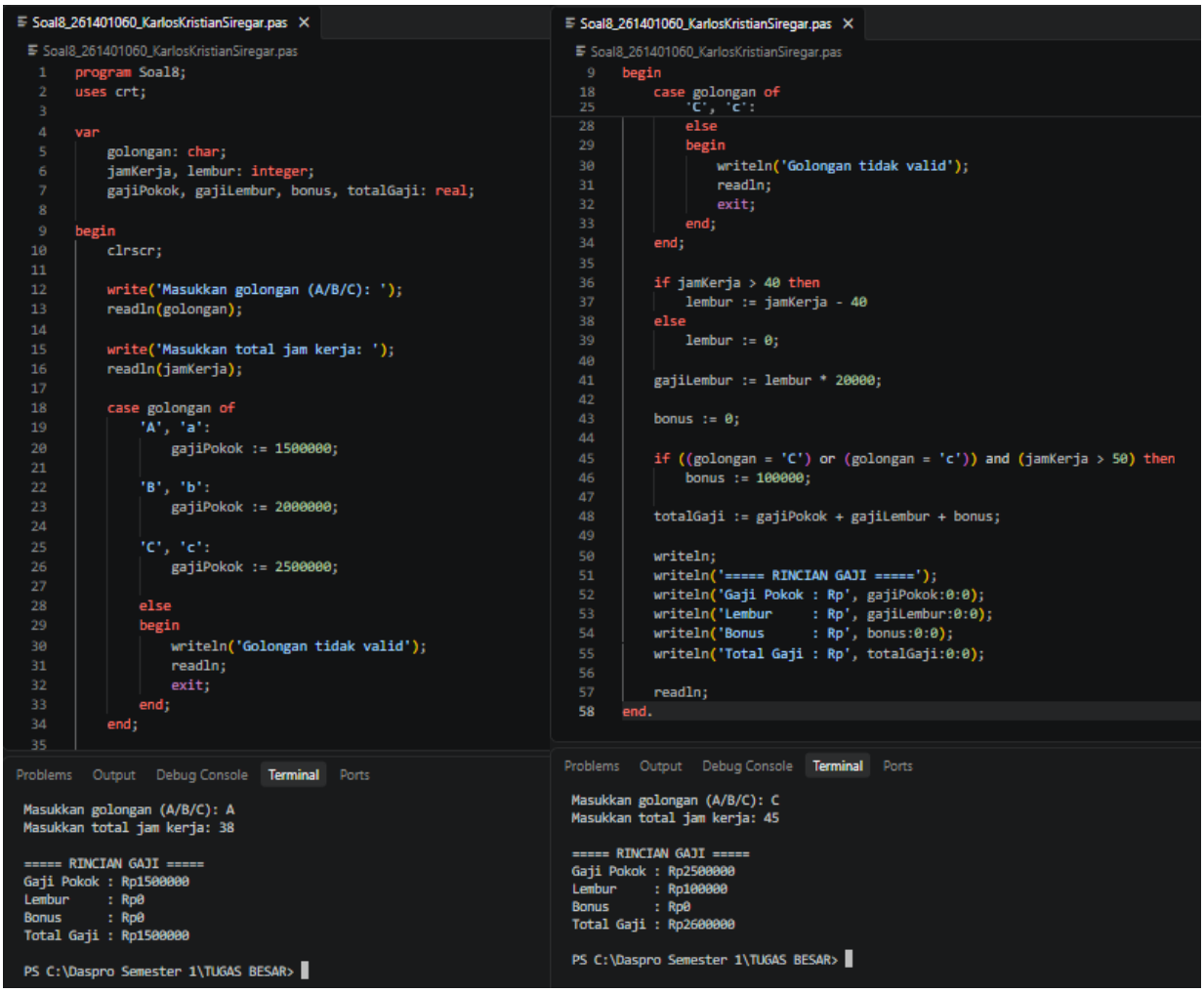
```
Soal7_261401060_KarlosKristianSiregar.pas
 8   begin
17       case kode of
44           begin
47               exit;
48           end;
49       end;
50
51       if (kode = 'M') or (kode = 'm') then
52       begin
53           if tarif > 30000 then
54               tarif := 30000;
55       end
56       else if (kode = 'K') or (kode = 'k') then
57       begin
```

Problems Output Debug Console **Terminal** Ports

```
Masukkan kode kendaraan (M/K/B): K
Masukkan lama parkir (jam): 8
Total tarif parkir = Rp9000

PS C:\Daspro Semester 1\TUGAS BESAR>
```

[Soal 8]



Soal8_261401060_KarlosKristianSiregar.pas

```
program Soal8;
uses crt;

var
    golongan: char;
    jamKerja, lembur: integer;
    gajiPokok, gajiLembur, bonus, totalGaji: real;

begin
    clrscr;

    write('Masukkan golongan (A/B/C): ');
    readln(golongan);

    write('Masukkan total jam kerja: ');
    readln(jamKerja);

    case golongan of
        'A', 'a':
            gajiPokok := 1500000;

        'B', 'b':
            gajiPokok := 2000000;

        'C', 'c':
            gajiPokok := 2500000;

        else
        begin
            writeln('Golongan tidak valid');
            readln;
            exit;
        end;
    end;
```

Problems Output Debug Console Terminal Ports

```
Masukkan golongan (A/B/C): A
Masukkan total jam kerja: 38

===== RINCIAN GAJI =====
Gaji Pokok : Rp1500000
Lembur     : Rp0
Bonus      : Rp0
Total Gaji : Rp1500000

PS C:\Daspro Semester 1\TUGAS BESAR>
```

Soal8_261401060_KarlosKristianSiregar.pas

```
begin
    case golongan of
        'C', 'c':
        else
        begin
            writeln('Golongan tidak valid');
            readln;
            exit;
        end;
    end;

    if jamKerja > 40 then
        lembur := jamKerja - 40
    else
        lembur := 0;

    gajiLembur := lembur * 20000;

    bonus := 0;

    if ((golongan = 'C') or (golongan = 'c')) and (jamKerja > 50) then
        bonus := 100000;

    totalGaji := gajiPokok + gajiLembur + bonus;

    writeln;
    writeln('===== RINCIAN GAJI =====');
    writeln('Gaji Pokok : Rp', gajiPokok:0:0);
    writeln('Lembur     : Rp', gajiLembur:0:0);
    writeln('Bonus      : Rp', bonus:0:0);
    writeln('Total Gaji : Rp', totalGaji:0:0);

    readln;
end.
```

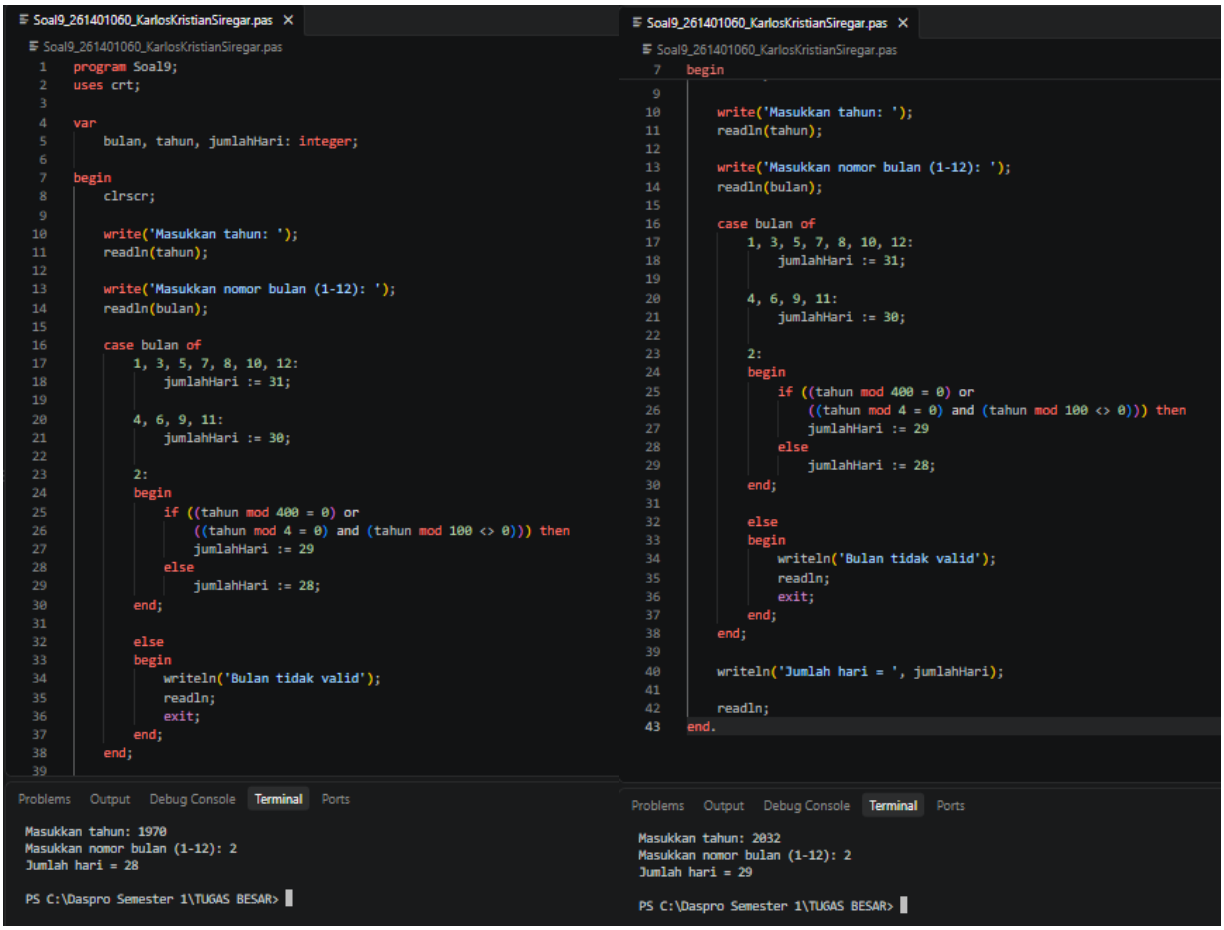
Problems Output Debug Console Terminal Ports

```
Masukkan golongan (A/B/C): C
Masukkan total jam kerja: 45

===== RINCIAN GAJI =====
Gaji Pokok : Rp2500000
Lembur     : Rp100000
Bonus      : Rp0
Total Gaji : Rp2600000

PS C:\Daspro Semester 1\TUGAS BESAR>
```

[Soal 9]



Soal9_261401060_KarlosKristianSiregar.pas

```
program Soal9;
uses crt;

var
    bulan, tahun, jumlahHari: integer;

begin
    clrscr;

    write('Masukkan tahun: ');
    readln(tahun);

    write('Masukkan nomor bulan (1-12): ');
    readln(bulan);

    case bulan of
        1, 3, 5, 7, 8, 10, 12:
            jumlahHari := 31;

        4, 6, 9, 11:
            jumlahHari := 30;

        2:
        begin
            if ((tahun mod 400 = 0) or
                ((tahun mod 4 = 0) and (tahun mod 100 <> 0))) then
                jumlahHari := 29
            else
                jumlahHari := 28;
        end;

        else
        begin
            writeln('Bulan tidak valid');
            readln;
            exit;
        end;
    end;
```

Problems Output Debug Console Terminal Ports

```
Masukkan tahun: 1970
Masukkan nomor bulan (1-12): 2
Jumlah hari = 28

PS C:\Daspro Semester 1\TUGAS BESAR>
```

Soal9_261401060_KarlosKristianSiregar.pas

```
begin

    write('Masukkan tahun: ');
    readln(tahun);

    write('Masukkan nomor bulan (1-12): ');
    readln(bulan);

    case bulan of
        1, 3, 5, 7, 8, 10, 12:
            jumlahHari := 31;

        4, 6, 9, 11:
            jumlahHari := 30;

        2:
        begin
            if ((tahun mod 400 = 0) or
                ((tahun mod 4 = 0) and (tahun mod 100 <> 0))) then
                jumlahHari := 29
            else
                jumlahHari := 28;
        end;

        else
        begin
            writeln('Bulan tidak valid');
            readln;
            exit;
        end;
    end;

    writeln('Jumlah hari = ', jumlahHari);

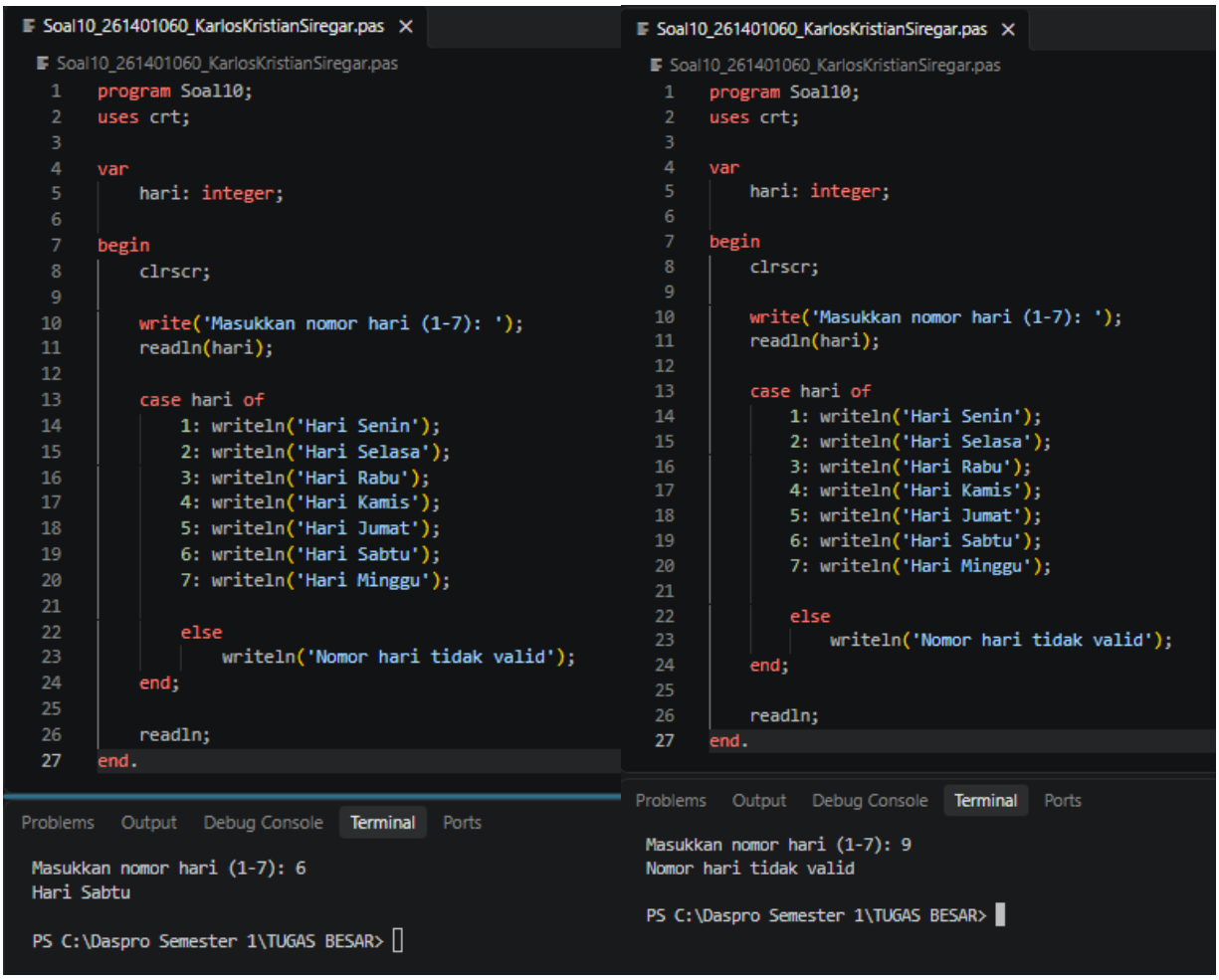
    readln;
end.
```

Problems Output Debug Console Terminal Ports

```
Masukkan tahun: 2032
Masukkan nomor bulan (1-12): 2
Jumlah hari = 29

PS C:\Daspro Semester 1\TUGAS BESAR>
```

[Soal 10]



Soal10_261401060_KarlosKristianSiregar.pas

```
program Soal10;
uses crt;

var
    hari: integer;

begin
    clrscr;

    write('Masukkan nomor hari (1-7): ');
    readln(hari);

    case hari of
        1: writeln('Hari Senin');
        2: writeln('Hari Selasa');
        3: writeln('Hari Rabu');
        4: writeln('Hari Kamis');
        5: writeln('Hari Jumat');
        6: writeln('Hari Sabtu');
        7: writeln('Hari Minggu');

        else
            writeln('Nomor hari tidak valid');
    end;

    readln;
end.
```

Problems Output Debug Console Terminal Ports

```
Masukkan nomor hari (1-7): 6
Hari Sabtu

PS C:\Daspro Semester 1\TUGAS BESAR>
```

Soal10_261401060_KarlosKristianSiregar.pas

```
program Soal10;
uses crt;

var
    hari: integer;

begin
    clrscr;

    write('Masukkan nomor hari (1-7): ');
    readln(hari);

    case hari of
        1: writeln('Hari Senin');
        2: writeln('Hari Selasa');
        3: writeln('Hari Rabu');
        4: writeln('Hari Kamis');
        5: writeln('Hari Jumat');
        6: writeln('Hari Sabtu');
        7: writeln('Hari Minggu');

        else
            writeln('Nomor hari tidak valid');
    end;

    readln;
end.
```

Problems Output Debug Console Terminal Ports

```
Masukkan nomor hari (1-7): 9
Nomor hari tidak valid

PS C:\Daspro Semester 1\TUGAS BESAR>
```